

AI Foundations & Developer Environment - Part 1

Q1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, understanding language, recognizing images, and making decisions. AI systems analyze data, identify patterns, and improve their performance over time.

Q2. Difference between AI, Machine Learning, Generative AI, and Agentic AI

AI: Broad field of creating intelligent machines.

Machine Learning: AI systems that learn from data.

Generative AI: AI that creates new content such as text, images, or code.

Agentic AI: AI that can plan actions and work toward goals with limited human guidance.

Q3. Explain any three open-source AI models available on Hugging Face

Llama 3: Open-source language model by Meta for chat, writing, and coding.

Mistral 7B: Efficient language model known for strong performance.

Stable Diffusion: Text-to-image model that generates images from prompts.

Q4. Difference between traditional programming and AI-assisted development

Traditional programming relies on manually written rules and logic. AI-assisted development uses AI tools to suggest code, automate tasks, detect bugs, and improve productivity.

Q5. Explain AI hallucinations with one example

An AI hallucination occurs when an AI generates information that sounds correct but is actually false. Example: claiming a winner of a future sports tournament that has not yet happened.

a

AI-GENERATED IMAGE:

